

Silt vs. Algae Differentiator (OFI)

- **Sensors Used:** Turbidity Sensor, Dual-Time pH Sensor, Water Temperature Sensor
- **Scope:** Per-Grid (Requires diurnal memory lookup).
- **Objective:** Differentiate non-biological mud/silt from photosynthetic microalgae.
- **Accuracy:** 80% to 85%
- **Mathematical Formula:**

$$OFI = \left(\frac{\text{Turbidity}}{\text{Baseline_Turbidity}} \right) \times \left(\frac{\text{pH}_{\text{midday}} - \text{pH}_{\text{morning}}}{\Delta T} \right)$$

- **Functional Logic:**
 - Read pH_morning at 08:00 hrs and store to grid coordinate memory GridMap[x][y].ph_morning.
 - Read pH_midday at 13:00 hrs at the same coordinate.
 - Compute normalized temperature delta (ΔT).
- **Threshold Outputs & Predictions:**
 - $OFI > 1.5$: Live Algae / Microalgae Dominance (Organic waste driver).
 - $0.3 \leq OFI \leq 1.5$: Mixed Organic & Inorganic suspension.
 - $OFI < 0.3$ (with High Turbidity): Inorganic Silt / Mud / Clay (Non-biological cloudiness).

Diurnal Algal Bloom Index (DABI)

- **Sensors Used:** Dual-Time pH Sensor, Turbidity Sensor
- **Scope:** Per-Grid (Requires diurnal memory lookup).
- **Objective:** Quantify algal bloom severity and assess nighttime oxygen drop risk.
- **Accuracy:** 80% to 85%
- **Mathematical Formula:**

$$DABI = (\text{pH}_{\text{midday}} - \text{pH}_{\text{morning}}) \times \sqrt{\text{Turbidity}}$$

- **Functional Logic:**
 - Read pH_morning at 08:00 hrs and pH_midday at 13:00 hrs from GridMap[x][y].
 - Sample current Turbidity at midday log.

- **Threshold Outputs & Predictions:**
 - $DABI > 8.0$: Active Algae/Cyanobacteria Bloom (Nighttime O_2 drop risk).
 - $2.0 \leq DABI \leq 8.0$: Stable biological activity.
 - $DABI < 2.0$ (with High Turbidity): Non-photosynthetic silt cloudiness.

Estimated Physical Secchi Depth (SD_{est})

- **Sensors Used:** Turbidity Sensor
- **Scope:** Real-Time Instantaneous / Per-Grid.
- **Objective:** Estimate light penetration depth without optical depth hardware.
- **Accuracy:** 75% to 80%
- **Mathematical Formula:**

$$SD_{est}(\text{meters}) = \frac{100}{\text{Turbidity} + 1}$$

- **Functional Logic:**
 - Continuous real-time calculation based on instantaneous sensor read.
 - Direct execution on micro-controller stream without historical memory dependency.
- **Threshold Outputs & Predictions:**
 - $SD_{est} < 0.3$ m (30 cm): Severe murky water (Bottom plants shaded, dying).
 - $0.3 \text{ m} \leq SD_{est} \leq 1.0$ m: Normal moderate clarity.
 - $SD_{est} > 1.0$ m: High transparency / Clear water column.

Organic Waste vs. Inorganic Salt Discriminator (TDS Dynamic Scaling / SOR)

- **Sensors Used:** TDS Sensor, Turbidity Sensor
- **Scope:** Real-Time Instantaneous.
- **Objective:** Differentiate non-biological mineral runoff from decaying organic waste.
- **Accuracy:** 80% to 85%
- **Mathematical Formula:**

$$SOR = \frac{TDS}{\text{Turbidity} + 1}$$

- **Functional Logic:**

- Compute inline dynamic scaling during active traversal.
- Evaluate real-time ratio of ionic dissolved solids against particulate cloudiness.
- **Threshold Outputs & Predictions:**
 - $SOR > 10.0$: Inorganic Salt Ingress / Fertilizer / Mineral Runoff.
 - $1.5 \leq SOR \leq 10.0$: Mixed Mineral & Particulate background.
 - $SOR < 1.5$: Heavy Organic Waste / Manure / Decaying Fish Feed.

Dissolved vs. Suspended Load Ratio (DSL_R)

- **Sensors Used:** TDS Sensor, Turbidity Sensor
- **Scope:** Real-Time Instantaneous.
- **Objective:** Classify water column turbidity to target chemical vs. mechanical cleaning actions.
- **Accuracy:** 80% to 85%
- **Mathematical Formula:**

$$DSL_R = \frac{TDS}{Turbidity + 1}$$

- **Functional Logic:**
 - Evaluate dynamic load ratio in streaming sensor pipeline.
 - Categorize dominance of dissolved ionic ions versus macro suspended solids.
- **Threshold Outputs & Predictions:**
 - $DSL_R > 10.0$: Dissolved Ion Dominance (Chemical/runoff treatment needed).
 - $1.5 \leq DSL_R \leq 10.0$: Balanced dissolved & suspended load.
 - $DSL_R < 1.5$: Particulate Suspended Dominance (Mechanical filtration needed).

Maximum Dissolved Oxygen Saturation Capacity (DO_{sat})

- **Sensors Used:** Water Temperature Sensor, TDS Sensor
- **Scope:** Real-Time Instantaneous.
- **Objective:** Calculate theoretical maximum dissolved oxygen storage capacity based on thermal and ionic loading.

- **Accuracy:** 85% to 90%
- **Mathematical Formula:**

$$DO_{\text{sat}}(\text{mg/L}) = 14.621 - 0.41022T + 0.007991T^2 - 0.000077774T^3 \\ - ((TDS \times 0.0007) \times 0.015)$$

- **Functional Logic:**
 - Sample continuous temperature (T in °C) and TDS (in ppm).
 - Calculate maximum physical saturation limit to evaluate biological stress and aeration needs.
- **Threshold Outputs & Predictions:**
 - $DO_{\text{sat}} < 6.5$ mg/L: Warm Water Stress Zone (Low O_2 capacity).
 - $6.5 \text{ mg/L} \leq DO_{\text{sat}} \leq 9.0$ mg/L: Moderate O_2 capacity.
 - $DO_{\text{sat}} > 9.0$ mg/L: Optimal Cool Water Oxygen Storage.